httperf

httperf is an open source, flexible and highly efficient Web server performance monitoring tool for generating various HTTP workloads. The three main characteristics of this tool are robustness (the ability to generate and sustain server overload), support for HTTP/1.1 and SSL protocols, and extensibility to new workload generators.

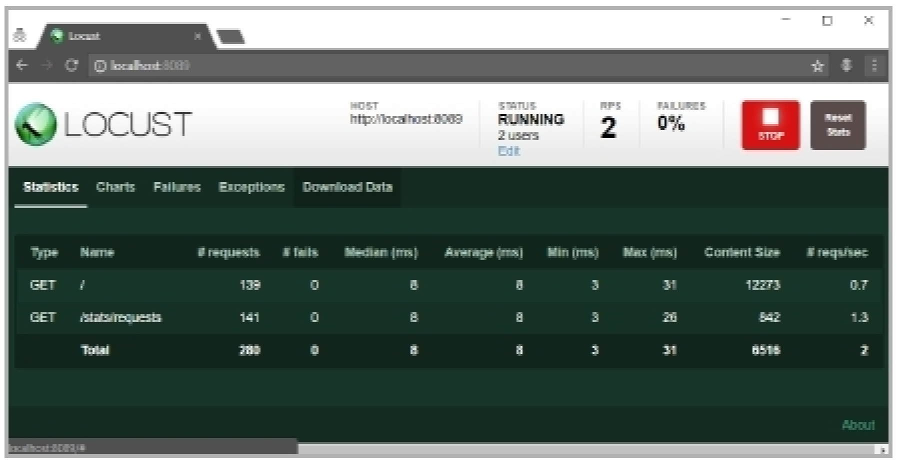

Figure 2: Locust.io Web UI

The main objective of this testing tool is to count the number of responses generated from a particular server. It generates the HTTP GET requests from the server, which helps in summarising the overall performance of the server. This was originally developed by David Mosberger and many others at HP. It is a Hewlett-Packard product.

Official website: *https://github.com/httperf/httperf*
Latest version: 0.9.0

Apache JMeter

Apache JMeter is an open source, Java based performance testing tool and can be integrated with a test plan. It can also be used as a load testing tool for analysing and measuring the performance of a variety of services, primarily, Web applications. Apache JMeter may be used to test performance both on static and dynamic resources, and on dynamic Web applications. It can be used to simulate a heavy load on a server, group of servers, network or object to test its strength or to analyse overall performance under different load types. It has 64-different plugins to speed up the process of creating and executing the JMeter Test plan.

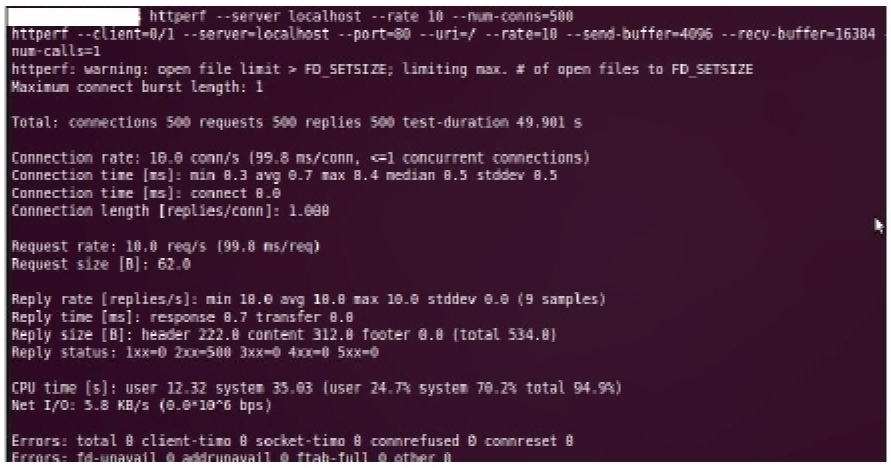

Figure 3: httperf

It is of great use in testing the functional performance of resources such as servlets, Perl scripts and Java objects.

Features:

- Can do the performance and load testing of many servers, protocols and applications like HTTP, HTTPS, SOAP, FTP, JDBC, LDAP, SMTP, POP3, IMAP, TCP, etc.
- Creates dynamic HTML reports, is fully portable and designed entirely in Java.
- Nice IDE for recording, building and debugging performance tests.
- Easy to use, as Groovy is the default programming language.

Official website: *https://jmeter.apache.org/*
Latest version: 5.0

Siege

Siege is an open source regression test and benchmark utility. It can stress-test a single URL with a user defined number of simulated users, or it can read many URLs into memory and stress-test them simultaneously. The program reports the total number of hits recorded, bytes transferred, response times, concurrency, and return status. Siege supports the HTTP/1.0 and 1.1 protocols, the GET and POST directives, cookies, transaction logging, and basic authentication. Its features are configurable on a per user basis. Siege allows you to stress test a Web server with 'n' number of users 't' number of times, where 'n' and 't' are defined by the user. It records the duration time of the test as well as the duration of each single transaction. It reports the number of transactions, elapsed time, bytes transferred, response times, transaction rate, concurrency and the number of times the server responded 'OK', i.e., status Code 200.

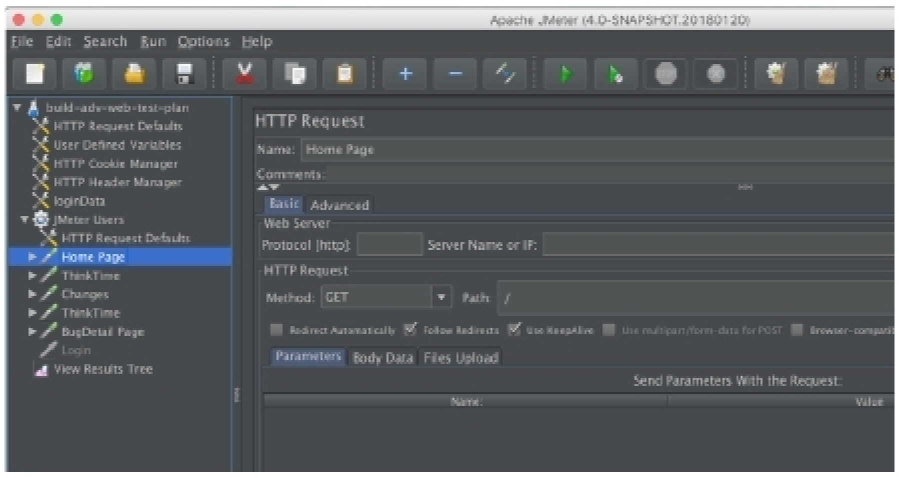

Figure 4: Apache JMeter

Features:

- Siege allows developers to place those programs under stress to allow a better understanding of the load that they can withstand.
- It supports basic authentication, cookies, the HTTP, HTTPS and FTP protocols.
- It allows its users to hit a server with a configurable number of simulated clients. Those clients place the server 'under siege'.